$\boxed{P3.17}$ $$\frac{d}{dt}\langle Q\rangle = \frac{i}{\hbar}\langle[\hat{H},\hat{Q}]\rangle + \langle\frac{\partial\hat{Q}}{\partial t}\rangle$$

(a) $\hat{Q}=1 \Rightarrow \frac{d}{dt}\langle 1\rangle = \frac{d}{dt}\langle\psi|\psi\rangle = 0$ because $[\hat{H},1]=0$.

This is conservation of normalization

(b) $\hat{Q}=\hat{H} \Rightarrow \frac{d}{dt}\langle\psi|\hat{H}\psi\rangle = \frac{i}{\hbar}\langle\psi|\underbrace{[\hat{H},\hat{H}]}_{\to 0}\psi\rangle = 0$

conservation of energy.

(c) $\hat{Q}=x \Rightarrow \frac{d}{dt}\langle x\rangle = \frac{i}{\hbar}\langle\psi|[\hat{H},\hat{x}]\psi\rangle$

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x) = \frac{\hat{p}^2}{2m} + V(x)$$

$$[\hat{H},\hat{x}]\psi = \left(\left[\frac{\hat{p}^2}{2m},\hat{x}\right] + \underbrace{[V(x),\hat{x}]}_{\emptyset}\right)\psi = \frac{1}{2m}\left(\hat{p}^2 x - x\hat{p}^2\right)\psi$$

Using P 3.13a : $[\hat{p}^2,x] = [\hat{p}\cdot\hat{p},x] = \hat{p}[\hat{p},x] + [\hat{p},x]\hat{p} =$

$$= \hat{p}(-i\hbar) + (-i\hbar)\hat{p} = -2i\hbar\hat{p}$$

$$\Rightarrow [\hat{H},x]\psi = \frac{-2i\hbar}{2m}\hat{p}\psi$$

$$\Rightarrow \frac{d}{dt}\langle x\rangle = \frac{i}{\hbar}\frac{-i\hbar}{m}\langle\psi|\hat{p}\psi\rangle = \langle\psi|\frac{\hat{p}}{m}\psi\rangle$$

$$= \langle\frac{\hat{p}}{m}\rangle$$

(It is quantum analog of $\frac{dx}{dt} = v = \frac{p}{m}$)

(d) $\hat{Q}=\hat{p} \Rightarrow \frac{d}{dt}\langle p\rangle = \frac{i}{\hbar}\langle[\frac{p^2}{2m}+V,p]\rangle =$

$$= \frac{i}{\hbar}\langle[V,p]\rangle = \frac{i}{\hbar}\langle i\hbar\frac{\partial V}{\partial x}\rangle = \langle -\frac{dV}{dx}\rangle$$

using P 3.13c

quantum analog of $\frac{dp}{dt} = F = -\frac{dV}{dx}$ 2nd Newton's law

Ehrenfest's theorem, eq. 1.3.8

$\boxed{P3.22}$ $|\alpha\rangle = i|1\rangle - 2|2\rangle - i|3\rangle$

$|\beta\rangle = i|1\rangle + 2|3\rangle$

(a) $\langle\alpha| = -i\langle 1| - 2\langle 2| + i\langle 3|$

$\langle\beta| = -i\langle 1| + 2\langle 3|$

(b) $\langle\alpha|\beta\rangle = \left(-i\langle 1| - 2\langle 2| + i\langle 3|\right)\left(i|1\rangle + 2|3\rangle\right) =$

$= -i\cdot i\,\langle 1|1\rangle + i\cdot 2\,\langle 3|3\rangle = 1+2i$

$\langle\beta|\alpha\rangle = \left(-i\langle 1| + 2\langle 3|\right)\left(i|1\rangle - 2|2\rangle - i|3\rangle\right)$

$= -i\cdot i\,\langle 1|1\rangle + 2(-i)\,\langle 3|3\rangle = 1-2i$

$\Rightarrow \langle\alpha|\beta\rangle = 1+2i = \langle\beta|\alpha\rangle^* = (1-2i)^*$ indeed.

(c) Operator $\hat{A} = |\alpha\rangle\langle\beta|$

Its (n,m) matrix element $A_{nm} = \langle n|\alpha\rangle\langle\beta|m\rangle$

where $|n\rangle, |m\rangle$ are from basis set $\{|1\rangle, |2\rangle, |3\rangle\}$.

$$\hat{A} \rightarrow \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 2i \\ 2i & 0 & -4 \\ -1 & 0 & -2i \end{pmatrix}$$

Need to use $\langle 1|\alpha\rangle = i$, $\langle 2|\alpha\rangle = -2$, $\langle 3|\alpha\rangle = -i$

$\langle\beta|1\rangle = -i$, $\langle\beta|2\rangle = 0$, $\langle\beta|3\rangle = 2$

Hermitian matrix if $(A^T)^* = A^\dagger = A$

In this case $A^\dagger = \begin{pmatrix} 1 & 2i & -1 \\ 0 & 0 & 0 \\ 2i & -4 & -2i \end{pmatrix}^* = \begin{pmatrix} 1 & -2i & -1 \\ 0 & 0 & 0 \\ -2i & -4 & 2i \end{pmatrix} \neq A$

<u>No</u>, $\hat{A}$ is not hermitian.

$\boxed{P3.23}$ $\hat{H} = E\left(|1\rangle\langle 1| - |2\rangle\langle 2| + |1\rangle\langle 2| + |2\rangle\langle 1|\right)$

Eigenvalue problem $\hat{H}|\psi\rangle = \omega|\psi\rangle$

where arbitrary $|\psi\rangle = \alpha|1\rangle + \beta|2\rangle$

$$\hat{H}\left(\alpha|1\rangle + \beta|2\rangle\right) = E\Big\{\alpha\left(|1\rangle\langle 1|1\rangle + |2\rangle\langle 1|1\rangle\right)$$

$$+\beta\left(-|2\rangle\langle 2|2\rangle + |1\rangle\langle 2|2\rangle\right)\Big\} = E\Big\{\alpha\left(|1\rangle + |2\rangle\right)$$

$$+\beta\left(-|2\rangle + |1\rangle\right)\Big\} = E\left\{(\alpha+\beta)|1\rangle + (\alpha-\beta)|2\rangle\right\} \Rightarrow$$

$$E(\alpha+\beta)|1\rangle + E(\alpha-\beta)|2\rangle = \omega\alpha|1\rangle + \omega\beta|2\rangle$$

$$\Rightarrow \begin{cases} E(\alpha+\beta) = \omega\alpha \\ E(\alpha-\beta) = \omega\beta \end{cases}$$

$$\beta = \frac{E}{\omega}(\alpha-\beta) \Rightarrow \beta\left(1+\frac{E}{\omega}\right) = \frac{E}{\omega}\alpha$$

$$\beta(\omega+E) = E\alpha$$

$$\beta = \frac{E}{\omega+E}\alpha$$

$$\Rightarrow E\left(\alpha + \frac{E}{\omega+E}\alpha\right) = \omega\alpha$$

$$\Rightarrow E(\omega+2E) = \omega(\omega+E)$$

$$E\omega + 2E^2 = \omega^2 + E\omega \Rightarrow \omega^2 = 2E^2 \Rightarrow \boxed{\omega = \pm\sqrt{2}E}$$

eigenvalues

$\underline{\omega = +\sqrt{2}E}$: eigenvector $\begin{cases} \alpha+\beta = \sqrt{2}\alpha \\ \alpha-\beta = \sqrt{2}\beta \end{cases} \Rightarrow \beta = \frac{\alpha}{1+\sqrt{2}} = (\sqrt{2}-1)\alpha$

$$|\psi_+\rangle = \alpha_+\left(|1\rangle + (\sqrt{2}-1)|2\rangle\right)$$

$\underline{\omega = -\sqrt{2}E}$: $\begin{cases} \alpha+\beta = -\sqrt{2}\alpha \\ \alpha-\beta = -\sqrt{2}\beta \end{cases} \Rightarrow \beta = \frac{\alpha}{1-\sqrt{2}} = -(\sqrt{2}+1)\alpha$

$$|\psi_-\rangle = \alpha_-\left(|1\rangle - (\sqrt{2}+1)|2\rangle\right)$$

To normalize, require $\langle \psi_\pm | \psi_\pm \rangle = 1$

$$\Rightarrow \quad |\alpha_+|^2 \left(1 + (\sqrt{2}-1)^2\right) = 1$$

$$|\alpha_+|^2 = \frac{1}{1+2-2\sqrt{2}+1} = \frac{1}{4-2\sqrt{2}}$$

and $|\alpha_-|^2 \left(1 + (\sqrt{2}+1)^2\right) = 1$

$$|\alpha_-|^2 = \frac{1}{1+2+2\sqrt{2}+1} = \frac{1}{4+2\sqrt{2}}$$

$$|\psi_+\rangle = \frac{1}{\sqrt{4-2\sqrt{2}}} \left(|1\rangle + (\sqrt{2}-1)\,|2\rangle\right)$$

$$|\psi_-\rangle = \frac{1}{\sqrt{4+2\sqrt{2}}} \left(|1\rangle - (\sqrt{2}+1)\,|2\rangle\right).$$

$$\text{Matrix } H = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} = \begin{pmatrix} E & E \\ E & -E \end{pmatrix} = E \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

A8 $\quad A = \begin{pmatrix} -1 & 1 & i \\ 2 & 0 & 3 \\ 2i & -2i & 2 \end{pmatrix} \qquad B = \begin{pmatrix} 2 & 0 & -i \\ 0 & 1 & 0 \\ i & 3 & 2 \end{pmatrix}$

(a) $A + B = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 3 \\ 3i & (3-2i) & 4 \end{pmatrix}$

(b) $AB = \begin{pmatrix} -3 & (1+3i) & 3i \\ (4+3i) & 9 & (6-2i) \\ 6i & (6-2i) & 6 \end{pmatrix}$

(c) $BA = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 3 \\ (6+3i) & -3i & 12 \end{pmatrix} \Rightarrow [A,B] = AB - BA = \begin{pmatrix} -3 & (1+3i) & 3i \\ (2+3i) & 9 & (3-2i) \\ (-6+3i) & (6+i) & -6 \end{pmatrix}$

(d) $\tilde{A} = \begin{pmatrix} -1 & 2 & 2i \\ 1 & 0 & -2i \\ i & 3 & 2 \end{pmatrix}$

(c) $A^* = \begin{pmatrix} -1 & 1 & -i \\ 2 & 0 & 3 \\ -2i & 2i & 2 \end{pmatrix}$

(f) $A^+ = (\tilde{A})^* = \begin{pmatrix} -1 & 2 & -2i \\ 1 & 0 & 2i \\ -i & 3 & 2 \end{pmatrix}$

(g) $\operatorname{Tr} B = 2+1+2 = 5$

(h) $\det B = 2\begin{vmatrix} 1 & 0 \\ 3 & 2 \end{vmatrix} - i\begin{vmatrix} 0 & 1 \\ i & 3 \end{vmatrix} = 2\cdot 2 - i(-i) = 3.$

(i) $B^{-1} = \frac{1}{3}\tilde{C}$, where C = cofactor matrix

$$C = \begin{pmatrix} 2 & 0 & -i \\ -3i & 3 & -6 \\ i & 0 & 2 \end{pmatrix} \Rightarrow \tilde{C} = \begin{pmatrix} 2 & -3i & i \\ 0 & 3 & 0 \\ -i & -6 & 2 \end{pmatrix}$$

$$B^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -3i & i \\ 0 & 3 & 0 \\ -i & -6 & 2 \end{pmatrix}$$

$$BB^{-1} = \begin{pmatrix} 2 & 0 & -i \\ 0 & 1 & 0 \\ i & 3 & 2 \end{pmatrix} \frac{1}{3}\begin{pmatrix} 2 & -3i & i \\ 0 & 3 & 0 \\ -i & -6 & 2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Check now $\det A = -1\begin{vmatrix} 0 & 3 \\ -2i & 2 \end{vmatrix} - 1\begin{vmatrix} 2 & 3 \\ 2i & 2 \end{vmatrix} + i\begin{vmatrix} 2 & 0 \\ 2i & -2i \end{vmatrix} =$

$= -6i - (4-6i) + i(-4i) = -4 + 4 = 0.$

$\Rightarrow$ A does not have inverse.

A18 $T = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

<u>Eigenvalues</u> from characteristic equation:

$$\begin{vmatrix} \cos\theta - \lambda & -\sin\theta \\ \sin\theta & \cos\theta - \lambda \end{vmatrix} = 0 = (\cos\theta - \lambda)^2 + \sin^2\theta = \cos^2\theta - 2\lambda\cos\theta + \lambda^2 + \sin^2\theta =$$

$$= 1 - 2\lambda\cos\theta + \lambda^2 = 0$$

$$\lambda = \cos\theta \pm \sqrt{\cos^2\theta - 1} = \cos\theta \pm i\sin\theta = e^{\pm i\theta}$$

$$\lambda_1 = e^{+i\theta}, \quad \lambda_2 = e^{-i\theta}$$

$\lambda_{1,2}$ are real only if $\theta = 0, \pi$.

<u>Eigenvectors</u>

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = e^{\pm i\theta} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\begin{cases} \cos\theta\,\alpha - \sin\theta\,\beta = e^{\pm i\theta}\alpha \\ \sin\theta\,\alpha + \cos\theta\,\beta = e^{\pm i\theta}\beta \end{cases} \Rightarrow \sin\theta\,\alpha = (e^{\pm i\theta} - \cos\theta)\beta =$$

$$= (\cos\theta \pm i\sin\theta - \cos\theta)\beta =$$

$$= \pm i\sin\theta\,\beta$$

$$\Rightarrow \alpha = \pm i\beta$$

$$|a^{(1)}\rangle = \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} i\beta_1 \\ \beta_1 \end{pmatrix} \Rightarrow \langle a^{(1)}|a^{(1)}\rangle = 1$$

$$\Rightarrow (-i\beta_1^*, \beta_1^*)\begin{pmatrix} i\beta_1 \\ \beta_1 \end{pmatrix} = 2|\beta_1|^2 = 1$$

$$\Rightarrow \beta_1 = \frac{1}{\sqrt{2}}$$

$$|a^{(1)}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} i \\ 1 \end{pmatrix}$$

$$\Rightarrow |a^{(2)}\rangle = \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} -i \\ 1 \end{pmatrix}$$

<u>Diagonalization</u> $\left(S^{-1}\right)_{ij} = \left(a^{(j)}\right)_i$

$$\Rightarrow S^{-1}_{11} = a^{(1)}_1 = \frac{i}{\sqrt{2}}, \quad S^{-1}_{12} = a^{(2)}_1 = \frac{-i}{\sqrt{2}} \quad ; \quad S^{-1}_{21} = a^{(1)}_2 = \frac{1}{\sqrt{2}} \quad ; \quad S^{-1}_{22} = a^{(2)}_2 = \frac{1}{\sqrt{2}}$$

$$\Rightarrow S^{-1} = \frac{1}{\sqrt{2}}\begin{pmatrix} i & -i \\ 1 & 1 \end{pmatrix} \quad \Rightarrow \quad S = \frac{1}{\det S^{-1}}\begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{pmatrix}^T = \frac{1}{\sqrt{2}}\begin{pmatrix} -i & i \\ 1 & 1 \end{pmatrix}^T$$

$$= \frac{1}{\sqrt{2}}\begin{pmatrix} -i & 1 \\ i & 1 \end{pmatrix}$$

$$\det S^{-1} = \begin{vmatrix} \frac{i}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{vmatrix} = \frac{i}{2} - \frac{-i}{2} = i$$

<u>T = transpose</u>

Then $$S\,T\,S^{-1} = \frac{1}{\sqrt{2}}\begin{pmatrix} -i & 1 \\ i & 1 \end{pmatrix}\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix} i & -i \\ 1 & 1 \end{pmatrix} =$$

$$= \frac{1}{2}\begin{pmatrix} -i & 1 \\ +i & 1 \end{pmatrix}\begin{pmatrix} i\cos\theta - \sin\theta & -i\cos\theta - \sin\theta \\ i\sin\theta + \cos\theta & -i\sin\theta + \cos\theta \end{pmatrix} =$$

$$= \frac{1}{2}\begin{pmatrix} -i & 1 \\ i & 1 \end{pmatrix}\begin{pmatrix} i e^{i\theta} & -i e^{-i\theta} \\ e^{i\theta} & e^{-i\theta} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} (-i\cdot i + 1)e^{i\theta} & ((-i)^2+1)e^{-i\theta} \\ (i^2+1)e^{i\theta} & (-i\cdot i+1)e^{-i\theta} \end{pmatrix}$$

$$= \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}$$ indeed.